Basanth Kumar Putta

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Objective

Associate Software Engineer with hands-on experience in AI Engineering, Data Engineering, and software development. Experienced in building RAG applications, AI-powered chatbots, document processing solutions, and data-driven applications. Currently contributing to client-facing data engineering initiatives involving SFTP pipeline enablement and data integration. Skilled in Python, SQL, LLMs, and developing scalable software solutions, with an interest in building reliable data pipelines and intelligent applications.

Experience

Associate Software Engineer *Pune (On-site), India Jan 2026 – Present*
circulants.ai

- Built and deployed a Retrieval-Augmented Generation (RAG) application for pharmaceutical knowledge retrieval by indexing and querying over **10,000+** domain-specific documents using LLMs and vector search.
- Developed an AI-powered chatbot for pharmaceutical query resolution, enabling semantic search and contextual responses from a knowledge base containing **10,000+** documents.
- Created a Clinical Study Report (CSR) generation platform that automated report creation by analyzing ADaM datasets with **30,000+** records and successfully processed **100+** clinical documents.
- Researched, evaluated, and benchmarked emerging multi-agent AI frameworks to identify scalable architectures for enterprise AI applications.
- Working on a client-facing Data Engineering project focused on enabling and operationalizing SFTP-based data pipelines for secure file transfer and data integration.
- Involved in pipeline enablement and coordination of data transfer requirements to support reliable data movement between client systems and downstream processing workflows.

AI and ML Intern *Hyderabad, India May 2024 – Jul 2024*
Cantilever Labs

- Built a Convolutional Neural Network (CNN) using TensorFlow/Keras for plant disease classification, achieving **90%** test accuracy on a dataset of **20,000+** images.
- Managed the complete machine learning lifecycle, including data preprocessing, augmentation, model training, evaluation, and deployment using Python and TensorFlow.
- Recognized as one of the top presenters among the intern cohort for effectively demonstrating the project's technical depth, model performance, and real-world applicability.

Projects

SFTP-Based Data Pipeline Enablement

- Contributing to the enablement of secure file transfer pipelines using SFTP to facilitate data exchange between client systems and downstream workflows.
- Supporting pipeline integration requirements to improve the reliability and accessibility of transferred data.
- **Tools Used:** Python, MobaXterm, FileZilla, AWS.

Clinical Study Report (CSR) Generation

- Automated the generation of structured Clinical Study Reports (CSRs) by analyzing Tables, Listings, and Figures (TLFs) using Large Language Models.
- Implemented an SDE-based context retrieval approach to incorporate historical study information, improving report consistency and contextual accuracy.
- Designed prompt engineering and document generation workflows to produce standardized, regulatory-compliant reports.

- **Tools Used:** TypeScript, LLMs, Generative AI, Prompt Engineering, SDE

OCR Engine for Intelligent Document Extraction *OCR*

- Constructed an AI-powered OCR engine to accurately extract structured and unstructured content from scanned documents and images.
- Building extraction pipelines for complex document elements including tables, lists, figures, and multi-column layouts while preserving document structure.
- Optimized preprocessing and post-processing techniques to improve OCR accuracy and downstream document understanding.
- **Tools Used:** Python, OCR, Deep Learning, Computer Vision, AI

Plant Disease Detection *plant.AI*

- Enhanced a web app to help farmers identify plant diseases using image-based predictions
- Tools Used: Python, AI, ML

VitaCare *VitaCare App*

- Built a healthcare assistant to digitize prescriptions and enable appointment booking
- Tools Used: Python, OCR

Skills

- **AI / ML:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Deep Learning, Computer Vision, Model Fine-tuning
- **Frameworks & Libraries:** LangChain, Hugging Face Transformers, TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy
- **Data Engineering:**SFTP, Data Pipelines, Data Integration, Data Processing, ETL/ELT Concepts, MobaXterm, AWS SFTP, FileZilla
- **Programming Languages:** Python, C, C++, JavaScript, SQL
- **Web Technologies:** HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, Bootstrap
- **Soft Skills:** Teamwork, Communication, Adaptability, Problem-Solving

Education

Geethanjali College of Engineering and Technology *Nov 2022 – Jun 2026*
Bachelor's Degree in Computer Science and Engineering - IoT

- GPA: 9.1/10.0

Certifications and Achievements

- DataBricks - Data engineer Associate
- DataBricks - GenAI Associate
- Third ranker in Engineering
- Merit Certificate - Academic Excellence
- IITH - Workshop
- Cisco Academy - Network Technician